

Model Answers ? Calculus Exam

DEMO FILE ? replace with the real document from the dashboard.

Q1. $\lim_{x \rightarrow 0} \sin(x)/x = 1$

Standard limit result.

Q2. $d/dx (x^3) = 3x^2$

Power rule: $d/dx (x^n) = n \cdot x^{(n-1)}$.

Q3. Integral of $2x \, dx = x^2 + C$

Since $d/dx (x^2) = 2x$.

Q4. Slope of tangent to $y = x^2$ at $x = 1$:

$dy/dx = 2x$, so $m = 2$.

Full worked solutions for every question are covered in the revision sessions.